

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

KIDS-PREK — Full working manuscript

Age guide: 4–6 years

Review prototype · not child-validated · not publication-ready

S01 — Spread S01 — The lamp that needs help (STORY)



figures/FIG-PREK-ME-TECH-S01.svg

Mira looks at a lamp. Bolt points to the switch, the cord, and the bulb. The screen of a phone nearby is only one part of another system.

Try it: Story hook: find three parts of one familiar device.

S02 — Spread S02 — What changes? (NOTICE)



figures/FIG-PREK-ME-TECH-S02.svg

When Mira presses the switch, light comes on. That press is an input. The light is an output.

Try it: Notice input then output once.

S03 — Spread S03 — System and part (NAME)



figures/FIG-PREK-ME-TECH-S03.svg

Bolt says the lamp is a system. A part is one piece that helps. Switch, cord, bulb — parts of one system.

Try it: Say system and part while pointing.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S04 — Spread S04 — Parts work together (CONNECT)



figures/FIG-PREK-ME-TECH-S04.svg

If one part is missing, the system may not work. Step lines up: power → switch → light.

Try it: Connect three part cards in order.

S05 — Spread S05 — What if the bulb is out? (PREDICT)



figures/FIG-PREK-ME-TECH-S05.svg

Mira predicts: if the bulb is out, the switch still clicks, but maybe no bright output.
What do you predict?

Try it: Predict before checking a safe toy lamp.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S06 — Spread S06 — Find the parts (TRY)



figures/FIG-PREK-ME-TECH-S06.svg

Try it with a caregiver: point to input, then point to output. Name one part of the system.

Try it: Hands-on part hunt on a safe device.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S07 — Spread S07 — Tell the story of parts (EXPLAIN)



figures/FIG-PREK-ME-TECH-S07.svg

Explain which parts work together. Mira listens. Mistakes are ok.

Try it: Explain without perfection pressure.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S08 — Spread S08 — Build a tiny system (MAKE)



figures/FIG-PREK-ME-TECH-S08.svg

Make a cardboard system: input card, part card, output card. Bolt helps tape the path.

Try it: Build a three-card system.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S09 — Spread S09 — Ask before opening (SAFE + FAIR)



figures/FIG-PREK-ME-TECH-S09.svg

Shield says: do not open real machines alone. Ask a trusted adult. Safe choices keep people and parts okay.

Try it: Choose the safer option: ask first.

KIDS FULL WORKING MANUSCRIPT
NOT CHILD-VALIDATED
NOT PUBLICATION-READY

S10 — Spread S10 — Teach the words (TEACH)



figures/FIG-PREK-ME-TECH-S10.svg

Teach your caregiver: system, part, input, output. You are the teacher now.

Try it: Teach-back four words with pointing.

S11 — Unit facilitator pointer



(See manuscript.)

S12 — Spread S01 — Two boxes on the table (STORY)



figures/FIG-PREK-INSIDE-S01.svg

Bolt shows Mira two boxes. One is Working Space — things we use right now. One is Keep-Box — things we save for later.

Try it: Story hook: now vs later.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S13 — Spread S02 — What stays? (NOTICE)



figures/FIG-PREK-INSIDE-S02.svg

Notice: a song playing now can stop. A picture saved in Keep-Box can still be there later.

Try it: Notice now versus later with a photo or drawing.

S14 — Spread S03 — Name the places (NAME)



figures/FIG-PREK-INSIDE-S03.svg

Working Space is for now. Keep-Box holds data for later. Data means information we keep.
Try it: Name Working Space and Keep-Box once.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S15 — Spread S04 — Same pattern (CONNECT)



figures/FIG-PREK-INSIDE-S04.svg

Step connects a pattern: make → use now → save to Keep-Box → find later.

Try it: Line up four step cards.

S16 — Spread S05 — If we do not save? (PREDICT)



figures/FIG-PREK-INSIDE-S05.svg

If we only use Working Space and never save, what do you predict happens later?

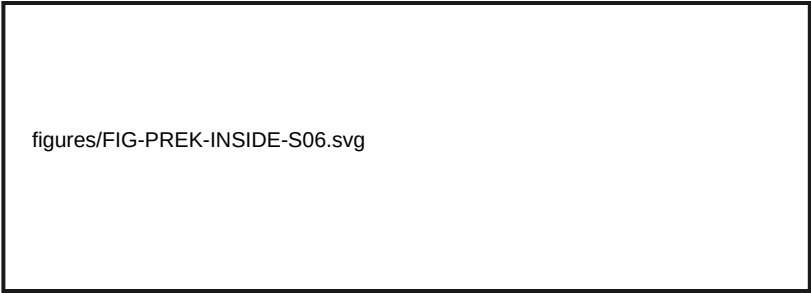
Try it: Predict before trying a drawing save.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S17 — Spread S06 — Save and find (TRY)

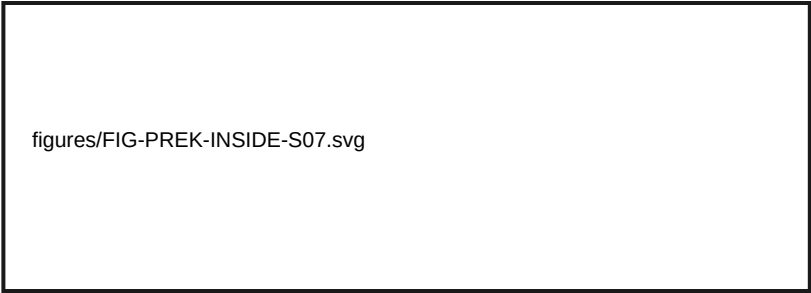


figures/FIG-PREK-INSIDE-S06.svg

Try it: draw a mark, put a copy in Keep-Box, then find it again with a caregiver.

Try it: Save-and-find paper activity.

S18 — Spread S07 — Explain the two places (EXPLAIN)

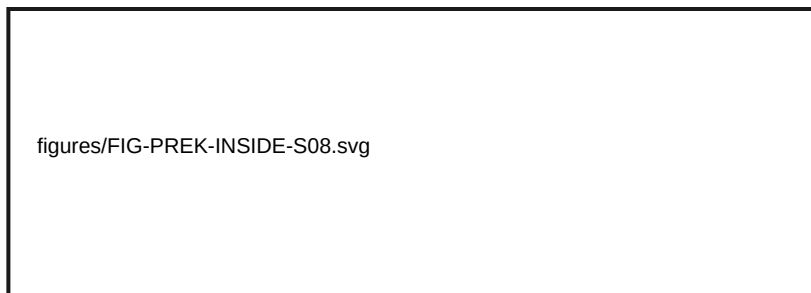


figures/FIG-PREK-INSIDE-S07.svg

Explain Working Space vs Keep-Box in your own words. Ping listens for the pattern.

Try it: Explain now vs later.

S19 — Spread S08 — Make sorting mats (MAKE)



Make two mats: Working Space and Keep-Box. Sort picture cards into each mat.

Try it: Hands-on sort.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S20 — Spread S09 — Who may see saved things? (SAFE + FAIR)



figures/FIG-PREK-INSIDE-S09.svg

Shield asks: some saved data is private. Ask a trusted adult before sharing Keep-Box photos.

Try it: Choose private vs share-with-adult.

S21 — Spread S10 — Teach now vs later (TEACH)



figures/FIG-PREK-INSIDE-S10.svg

Teach your caregiver the difference between Working Space and Keep-Box.

Try it: Teach-back with the two mats.

S22 — Unit facilitator pointer



(See manuscript.)

S23 — Spread S01 — Step brings cards (STORY)



figures/FIG-PREK-INSTRUCTIONS-S01.svg

Step brings three cards. An instruction tells what to do. A sequence is instructions in order.

Try it: Story hook: ordered cards.

S24 — Spread S02 — First, next, last (NOTICE)



figures/FIG-PREK-INSTRUCTIONS-S02.svg

Notice first, next, last on a snack helper: 1 wash hands 2 get cup 3 pour water.

Try it: Notice three ordered steps.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S25 — Spread S03 — Instruction and sequence (NAME)



figures/FIG-PREK-INSTRUCTIONS-S03.svg

Name it: each card is an instruction. Together in order they make a sequence — a simple algorithm.

Try it: Say instruction and sequence once.

S26 — Spread S04 — Input to output (CONNECT)



figures/FIG-PREK-INSTRUCTIONS-S04.svg

Connect: input (press) → instructions (steps) → output (what you see). Apps follow rules like this.

Try it: Connect IPO cards.

S27 — Spread S05 — Wrong order? (PREDICT)



figures/FIG-PREK-INSTRUCTIONS-S05.svg

If we pour water before getting the cup, what do you predict? Step smiles — sequences can be fixed.

Try it: Predict a mixed-up sequence.

S28 — Spread S06 — Run the steps (TRY)



figures/FIG-PREK-INSTRUCTIONS-S06.svg

Try a three-step sequence with paper cards. Follow first, next, last. Stop anytime.

Try it: Run a caregiver-safe sequence.

S29 — Spread S07 — Explain your sequence (EXPLAIN)



figures/FIG-PREK-INSTRUCTIONS-S07.svg

Explain your sequence. Why does order matter? Mira listens.

Try it: Explain without perfection pressure.

KIDS FULL WORKING MANUSCRIPT
NOT CHILD-VALIDATED
NOT PUBLICATION-READY

S30 — Spread S08 — Build-a-sequence (MAKE)



figures/FIG-PREK-INSTRUCTIONS-S08.svg

Build cards: Input → Steps → Output. Add a Fix card if something goes wrong.

Try it: Hands-on sequence build.

S31 — Spread S09 — Fair turns with steps (SAFE + FAIR)



figures/FIG-PREK-INSTRUCTIONS-S09.svg

Shield says take turns giving instructions. Fair means everyone can try the sequence.

Try it: Choose fair turn-taking.

S32 — Spread S10 — Teach three steps (TEACH)



figures/FIG-PREK-INSTRUCTIONS-S10.svg

Teach your caregiver a three-step sequence. You are the teacher now.

Try it: Teach-back three steps.

S33 — Unit facilitator pointer



(See manuscript.)

S34 — Spread S01 — Ping draws a path (STORY)



figures/FIG-PREK-MESSAGES-S01.svg

Ping draws a dotted path from Mira to a helper across the room. A message can travel here to there.

Try it: Story hook: here to there.

S35 — Spread S02 — Local or far? (NOTICE)



figures/FIG-PREK-MESSAGES-S02.svg

Notice: sometimes the answer is already here. Sometimes a message must ask far away.

Try it: Notice local vs far ask.

S36 — Spread S03 — Message and path (NAME)



figures/FIG-PREK-MESSAGES-S03.svg

Name it: a message is information that travels. A path is the way it goes. Rules tell how to send it.

Try it: Say message and path once.

S37 — Spread S04 — Rules for the path (CONNECT)



figures/FIG-PREK-MESSAGES-S04.svg

Connect the pattern: start → follow path → arrive → reply. Rules keep the path fair and clear.

Try it: Connect four path cards.

S38 — Spread S05 — If the path is blocked? (PREDICT)



figures/FIG-PREK-MESSAGES-S05.svg

If a door is closed on the path, what do you predict? Maybe wait, or try another path later.

Try it: Predict a blocked path.

S39 — Spread S06 — Draw Ping's path (TRY)



figures/FIG-PREK-MESSAGES-S06.svg

Try it: draw a simple out-and-back message path on paper. Ping rides the dots.

Try it: Draw out-and-back path.

S40 — Spread S07 — Explain the path (EXPLAIN)



figures/FIG-PREK-MESSAGES-S07.svg

Explain which path you drew and why. Mistakes are ok.

Try it: Explain the path.

S41 — Spread S08 — Message relay (MAKE)



figures/FIG-PREK-MESSAGES-S08.svg

Make a paper-note relay: write a kind mark, follow the path, deliver. Add a Rule card: ask first.

Try it: Hands-on relay with rule.

S42 — Spread S09 — Who may get the message? (SAFE + FAIR)



figures/FIG-PREK-MESSAGES-S09.svg

Shield says: do we send to a stranger? No — ask a trusted adult. Safe and private choices matter.

Try it: Choose safer recipient.

S43 — Spread S10 — Teach the path (TEACH)



figures/FIG-PREK-MESSAGES-S10.svg

Teach your caregiver the out-and-back path and one rule.

Try it: Teach-back path + rule.

S44 — Unit facilitator pointer



diagram

(See manuscript.)

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S45 — Spread S01 — A drawing to keep (STORY)



figures/FIG-PREK-DATA-S01.svg

Mira makes a drawing. Bolt says we can save it — keep the data for later — then find it again.

Try it: Story hook: create, save, find.

S46 — Spread S02 — Create, save, find (NOTICE)

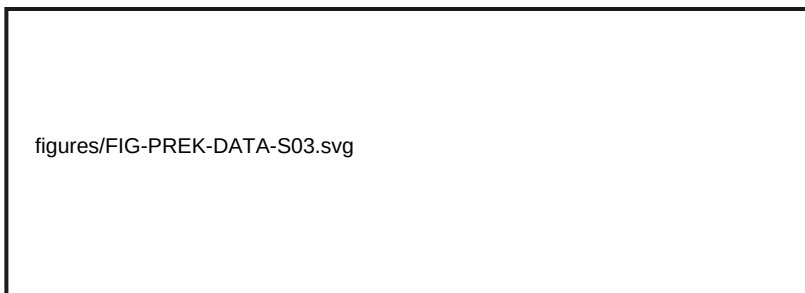


figures/FIG-PREK-DATA-S02.svg

Notice the pattern: create → save → find. Missing a step changes what happens later.

Try it: Notice the three-step pattern.

S47 — Spread S03 — Name the data (NAME)



Name it: the drawing is data. A name or picture label helps us find it in the Keep-Box.

Try it: Give a drawing a simple name label.

S48 — Spread S04 — Same idea in devices (CONNECT)



figures/FIG-PREK-DATA-S04.svg

Connect: phones and computers also create, save, and find. The pattern is the same, even if insides differ.

Try it: Connect paper pattern to a caregiver-shown save.

S49 — Spread S05 — If we skip save? (PREDICT)



figures/FIG-PREK-DATA-S05.svg

If we create but do not save, what do you predict when we look later?

Try it: Predict missing save.

KIDS FULL WORKING MANUSCRIPT
NOT CHILD-VALIDATED
NOT PUBLICATION-READY

S50 — Spread S06 — Save and find again (TRY)



figures/FIG-PREK-DATA-S06.svg

Try it: create a mark, save it in an envelope, then find it. Caregiver helps.

Try it: Create-save-find try.

S51 — Spread S07 — Explain the pattern (EXPLAIN)



figures/FIG-PREK-DATA-S07.svg

Explain create, save, find. Step listens for the pattern.

Try it: Explain the three words.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S52 — Spread S08 — Make a find box (MAKE)



figures/FIG-PREK-DATA-S08.svg

Make a Find Box with labeled envelopes. Put drawings in. Practice finding by label.

Try it: Build labeled find box.

S53 — Spread S09 — Private data (SAFE + FAIR)



figures/FIG-PREK-DATA-S09.svg

Shield says some data is private — not for every screen or stranger. Ask before sharing.

Try it: Choose private vs share-with-adult.

S54 — Spread S10 — Teach create-save-find (TEACH)



figures/FIG-PREK-DATA-S10.svg

Teach your caregiver the create → save → find pattern.

Try it: Teach-back the pattern.

S55 — Unit facilitator pointer



(See manuscript.)

S56 — Spread S01 — Shield's two doors (STORY)



figures/FIG-PREK-SAFE-S01.svg

Shield shows two doors: Safe Choice and Pause-to-Ask. Mira practices choosing before sharing a screen.

Try it: Story hook: choose before share.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S57 — Spread S02 — Who can see? (NOTICE)



figures/FIG-PREK-SAFE-S02.svg

Notice who can see: family, friend, or stranger? Private means not for everyone.

Try it: Notice audience options.

S58 — Spread S03 — Safe and private (NAME)



figures/FIG-PREK-SAFE-S03.svg

Name it: safe means people stay okay. Private means some information stays with trusted people.

Try it: Say safe and private once.

S59 — Spread S04 — Fair includes helpers (CONNECT)



figures/FIG-PREK-SAFE-S04.svg

Connect: fair means friends who need different tools can still join. Accessibility is part of fairness.

Try it: Connect fair with including helpers.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S60 — Spread S05 — Share with a stranger? (PREDICT)



figures/FIG-PREK-SAFE-S05.svg

If someone unknown asks to see the screen, what do you predict is safer?

Try it: Predict safer choice.

S61 — Spread S06 — Practice the choice (TRY)



figures/FIG-PREK-SAFE-S06.svg

Try choice cards: Share with caregiver / Ask first / Not with a stranger. Pick the safer path.

Try it: Choice-card practice.

S62 — Spread S07 — Explain your choice (EXPLAIN)

figures/FIG-PREK-SAFE-S07.svg

Explain why you chose safe or private. Shield listens. Mistakes are ok.

Try it: Explain the choice.

S63 — Spread S08 — Make a pause sign (MAKE)



figures/FIG-PREK-SAFE-S08.svg

Make a Pause-and-Ask sign for the family device corner. Practice holding it up.

Try it: Craft pause sign.

S64 — Spread S09 — Stop is always ok (SAFE + FAIR)



figures/FIG-PREK-SAFE-S09.svg

Shield says stop is always ok. Safe, private, and fair choices can be made slowly.

Try it: Practice an easy stop.

S65 — Spread S10 — Teach safe or private (TEACH)



figures/FIG-PREK-SAFE-S10.svg

Teach your caregiver one safe choice and one private choice.

Try it: Teach-back two choices.

S66 — Unit facilitator pointer



diagram

(See manuscript.)

S67 — Spread S01 — Plan on the table (STORY)



Step and Bolt help Mira plan a sequence, try it, then fix it. First tries do not need to be perfect.

Try it: Story hook: plan → try → fix.

S68 — Spread S02 — Two versions (NOTICE)



figures/FIG-PREK-BUILD-S02.svg

Notice Version 1 and Version 2 of the same plan. Saving versions helps us go back.

Try it: Notice two versions side by side.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S69 — Spread S03 — Test means check (NAME)



figures/FIG-PREK-BUILD-S03.svg

Name it: a test means we check if the sequence works. If not, we fix and try again.

Try it: Say test once with a check gesture.

S70 — Spread S04 — Plan, try, fix (CONNECT)



figures/FIG-PREK-BUILD-S04.svg

Connect the pattern: plan → try → test → fix. That is how builders improve a sequence.

Try it: Connect four builder cards.

S71 — Spread S05 — What will Version 2 fix? (PREDICT)



figures/FIG-PREK-BUILD-S05.svg

Look at a wobbly Version 1. What do you predict Version 2 should change?

Try it: Predict a fix.

S72 — Spread S06 — Try and test (TRY)



figures/FIG-PREK-BUILD-S06.svg

Try your sequence. Test it once. If it wobbles, fix one thing. Stop anytime.

Try it: Try-test-fix loop once.

S73 — Spread S07 — Explain what changed (EXPLAIN)



figures/FIG-PREK-BUILD-S07.svg

Explain what changed from Version 1 to Version 2. Ping listens.

Try it: Explain the fix.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S74 — Spread S08 — Keep both versions (MAKE)



figures/FIG-PREK-BUILD-S08.svg

Make Version 1 and Version 2 cards. Keep both. Builders save versions so they can go back.

Try it: Keep two version artifacts.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S75 — Spread S09 — Share kindly (SAFE + FAIR)



figures/FIG-PREK-BUILD-S09.svg

Shield says share your build only with trusted people. Fair feedback is kind and specific.

Try it: Practice kind share words.

KIDS FULL WORKING MANUSCRIPT

NOT CHILD-VALIDATED

NOT PUBLICATION-READY

S76 — Spread S10 — Teach plan-try-fix (TEACH)



figures/FIG-PREK-BUILD-S10.svg

Teach your caregiver plan → try → test → fix. You are the teacher now.

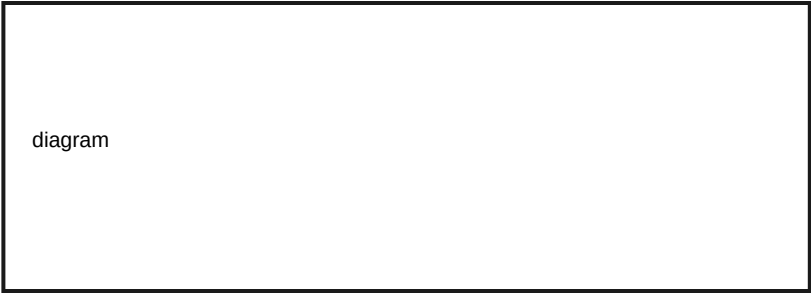
Try it: Teach-back the builder pattern.

S77 — Unit facilitator pointer



(See manuscript.)

S78 — Glossary (caregiver-facing terms used with children)



diagram

(See manuscript.)

S79 — Caregiver / educator extensions



(See manuscript.)

S80 — Standards / evidence appendix



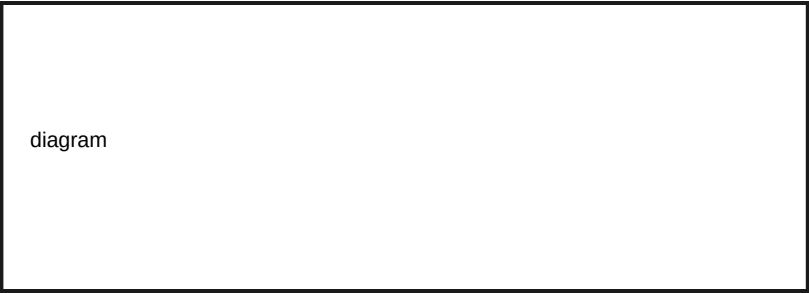
(See manuscript.)

S81 — Accessibility alternatives



(See manuscript.)

S82 — Provenance



(See manuscript.)